

sol4_fixer_agent_sdk

The working Module 4 loop. Issue 120.

`query(), not ClaudeSDKClient.permission_mode: dontAsk.` Merge is never a tool.

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Layout

loop.py	summarize_failure + repair_until_green wrappers
fixer.py	the while loop
doers.py	none / reference / cli / SDK backend
gates.py	identical copy to sol3
contract.py	Taskfile + junit + coverage
roles.py	dontAsk, one PreToolUse hook, deny tests/**
tests/	hook deny/allow, same-signature escalate

No `unattended.py`. Durable `.harness/state.json` and `exit 0/2/1` lived there and were not copied. This CLI returns `0` if green else `1`. Name that drift.

Learning objectives

- Walk `fixer.run`
- Set `permission_mode: dontAsk` because nobody is in the chair
- Deny `tests/**` on the code implementer
- Stash before `--branch broken-pr`
- Leave a comment when the loop gives up

Starting architecture

```
Trigger → fixer.run
        contract.run("test")
        green? PASS
        never ran? ESCALATE
        same failing ids twice? ESCALATE + comment
        else doer repairs inside app/**
→ .harness/last-fixer.json
→ Human merge
```

Five unattended lines

```
permission_mode: dontAsk      # never prompts, and fails closed
PreToolUse deny tests/**     # cannot weaken a test
maxTurns=12                   # camelCase. max_turns= is a TypeError
tests after every turn = pytest # not a claim
Merge is never a tool
```

This port used `acceptEdits` and that was the bug. Both modes never prompt. `acceptEdits` **auto-accepts every edit before** the allow list is read, so the hook is the only fence, and the hook fails open on a typo. `dontAsk` denies anything not pre-approved, which makes `allowed_tools` a real second gate.

"Nobody is in the chair" is the argument for `dontAsk`, not against it.

failure_summary

```
def failure_summary(run_result) -> str:
    failed = sorted(run_result.junit.failed_ids)
    lines = [f"{len(failed)} failing: {'', '}.join(failed[:5])}" if failed else ["the suite is red"]
    error = ERROR_IN_OUTPUT.search(run_result.output or "")
    if error:
        lines.append(error.group(0).strip()[:200])
    return "\n".join(lines)
```

Doers

	SPEC	BEHAVIOR
none		writes nothing. Loop still reports the truth
reference		copies <code>known-good</code> inside <code>WriteScope</code>
sdk		<code>AgentSdkBackend</code> wrapping <code>query()</code>
<code>claude / codex / grok</code>		<code>CliBackend</code>

Reference skipping `tests/**` is the whole point.

Commands

```
cd solutions/sol4_fixer_agent_sdk
python3 -m pytest tests -q
python3 loop.py --table-only
# after stash:
python3 loop.py --repo ../../work/northwind-field-crm --branch broken-pr --doer none
python3 loop.py --repo ../../work/northwind-field-crm --branch broken-pr --doer reference
```

Expected `--doer none`

```
attempt 1: 1 failing -> retry
attempt 2: 1 failing -> escalate
gate: escalate
The fixer gave up.
A human should take this one.
```

Tests

TEST

test_hook_denies_test_write

test_hook_allows_app_write

test_the_permission_mode_denies_rather_than_prompts

test_same_failing_ids_escalate

test_no_loops_import

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ASSERTS

Write tests/test_x.py → deny

Write app/main.py → {}

permission_mode == "dontAsk"

same signature → ESCALATE

no from loops

Troubleshooting

SYMPTOM	CAUSE	FIX
Dirty tree	Lab 2 leftover	stash
Green on <code>none</code>	missing signature stop	<code>gates.decide</code>
Edited a test	hook fail-open	full deny envelope
Exit 1 on escalate	unattended.py missing	named drift, not a silent bug

Recap

Same graph, nobody at the keyboard. Reference doer still bound by WriteScope. Human owns merge.

Rebuild `unattended.py` in this folder if you need exit 2 for escalate.

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Prerequisites

```
cd solutions/sol4_fixer_agent_sdk  
python3 -m pytest tests -q  
git -C ../../work/northwind-field-crm stash --include-untracked
```

--doer sdk **needs** claude-agent-sdk. --doer none **and** --doer reference **do not**.

`repair_until_green` **wrapper**

```
def repair_until_green(contract, budget: int = 3, doer: str = "reference") -> dict:
    if doer == "sdk":
        doer = backend(contract)
    return fixer.run(repo=contract.repo, budget=budget, doer=doer, research_backend=None)
```

Saturday types a body. This folder delegates to `fixer.run`.

Final checklist

- `[]` pytest green without a key
- `[]` table: judge `no`, coder denied `tests/**`
- `[]` `permission_mode == "dontAsk"`
- `[]` `--doer none` on broken-pr escalates with a comment
- `[]` merge is not in any tool list

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